



Society of Petroleum Engineers

SPE-229602-MS

Formation Damage Decoded: A Lab-Driven Workflow for Conventional Reservoirs. A Proven Laboratory Sequence to Identify and Mitigate Formation Damage

Ernesto Daniel Cabello and Juan Ignacio Cresta, LCV S.R.L, Buenos Aires, Argentina; Patricio Zamora, SLB, Quito, Ecuador

Copyright 2025, Society of Petroleum Engineers DOI [10.2118/229602-MS](https://doi.org/10.2118/229602-MS)

This paper was prepared for presentation at the ADIPEC held in Abu Dhabi, UAE, 3 – 6 November 2025.

This paper was selected for presentation by an SPE program committee following review of information contained in an abstract submitted by the author(s). Contents of the paper have not been reviewed by the Society of Petroleum Engineers and are subject to correction by the author(s). The material does not necessarily reflect any position of the Society of Petroleum Engineers, its officers, or members. Electronic reproduction, distribution, or storage of any part of this paper without the written consent of the Society of Petroleum Engineers is prohibited. Permission to reproduce in print is restricted to an abstract of not more than 300 words; illustrations may not be copied. The abstract must contain conspicuous acknowledgment of SPE copyright.

Formation damage is a critical concern in the efficient development of conventional oil and gas reservoirs, as any alteration of the target rock's permeability can result in substantial productivity losses and elevated operational and economic costs. This phenomenon encompasses a set of physical, chemical, and mechanical processes that occur during the drilling, completion, stimulation, and even production phases, with effects that may manifest both in the short and long term. Quantitatively, permeability losses associated with formation damage may range from 10% to 50%—or even higher, with field experience reporting up to 80%—of the reservoir's original capacity. The impact is particularly severe in mature fields undergoing secondary or tertiary recovery programs, as well as in wells drilled or completed using water injection or other fluid-based treatments ("products").

Formation Damage: From a scientific perspective, formation damage is defined as any adverse change in the petrophysical properties of the porous medium—particularly effective porosity, pore size distribution, and hydraulic conductivity—caused by fluid invasion, precipitate formation, or fine particle mobilization. These processes can be grouped into three main categories:

INTRODUCTION

Formation damage is a critical concern in the efficient development of conventional oil and gas reservoirs, as any alteration of the target rock's permeability can result in substantial productivity losses and elevated operational and economic costs. This phenomenon encompasses a set of physical, chemical, and mechanical processes that occur during the drilling, completion, stimulation, and even production phases, with effects that may manifest both in the short and long term. Quantitatively, permeability losses associated with formation damage may range from 10% to 50%—or even higher, with field experience reporting up to 80%—of the reservoir's original capacity. The impact is particularly severe in mature fields undergoing secondary or tertiary recovery programs, as well as in wells drilled or completed using water injection or other fluid-based treatments ("products").

Formation Damage

From a scientific perspective, formation damage is defined as any adverse change in the petrophysical properties of the porous medium—particularly effective porosity, pore size distribution, and hydraulic conductivity—caused by fluid invasion, precipitate formation, or fine particle mobilization. These processes can be grouped into three main categories:

1. **Fines migration and mechanical blockage:** Conventional sedimentary rocks often contain small clay and quartz particles that, when mobilized by filtrate flow or transient pressure changes, can migrate and accumulate within narrow pore channels. The resulting blockage of pores and pore throats leads to a drastic reduction in permeability.
2. **Hydrochemical interaction and solid precipitation:** The injection of fluids with chemical compositions different from the formation water can trigger dissolution or precipitation reactions within the pore network. For example, mixing low-salinity waters with carbonate-rich rocks may induce controlled calcite dissolution followed by reprecipitation in other areas. Similarly, ionic incompatibility between calcium and sulfate ions favors the formation of calcium sulfate (gypsum), strontium sulfate (celestite), or barium sulfate (barite), which can crystallize and block flow pathways.
3. **Wettability alteration and interfacial tension changes:** The modification of the rock minerals' affinity for water or oil—i.e., their wettability—affects spontaneous imbibition mechanisms and oil phase entrapment. Completion or fracturing fluids containing aggressive surfactants can alter phase distribution at the rock—fluid interface, shifting critical saturation and relative permeability endpoints, potentially with negative consequences.

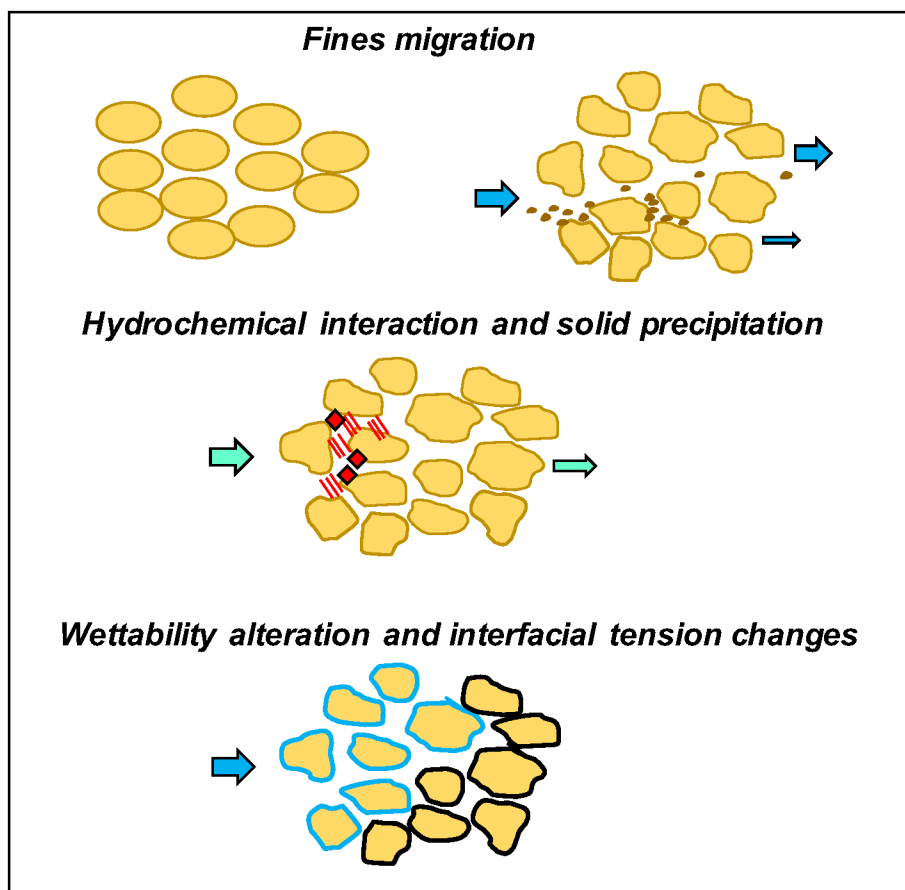


Figure 1—Examples of the main categories of damage described.

To systematically address these challenges, this article proposes a robust, reproducible, and cost-effective laboratory workflow specifically designed to assess and diagnose formation damage in conventional reservoirs. The methodology is supported by a detailed case study conducted in one of the main reservoirs of the Napo Formation, located in the Oriente Basin, Ecuador—a mature asset with a proven history of water injection as a secondary recovery method.

The primary objective is to present an effective workflow for the integrated study of conventional hydrocarbon reservoirs, taking into account potential formation damage from the earliest stages of reservoir evaluation, through laboratory testing on core samples.

Geological Context

The Napo and Hollm Formations host the main reservoirs of the Oriente Basin, including the Hollfn, T, U, and M1 reservoirs, along with marginal reservoirs such as the M2 Sandstone and the A, B, M2, and M1 Limestones. The Lower T and Lower U reservoirs display the best petrophysical properties, associated with fluvial and estuarine facies developed over erosive surfaces generated by sea-level falls (Baby et al., 1999).

The Napo Formation was deposited during the mid-to-late Cretaceous (late Albian to Campanian) within a pre-Andean ramp system influenced by eustatic fluctuations and tectonic subsidence of the foreland basin. It is divided into three members:

- **Lower Napo**, dominated by dark shales and siltstones rich in organic matter, deposited during maximum marine flooding stages.
- **Middle Napo**, consisting of interbedded bioturbated glauconitic sandstones and shales, representing shallow ramp systems under tidal influence. Its basal sandstone unit, the *Lower U Sandstone*, is the most productive reservoir due to its high intergranular porosity and high permeability.
- **Upper Napo**, characterized by sandy beds interbedded with shales and calcareous horizons, formed during high-energy regressive phases and progradation of inner shelf facies.

The combination of fluvio-estuarine and marine deposits, linked to complete regression-transgression-progradation cycles, accounts for the lithological variability and reservoir distribution. Structurally, the Napo Formation is hosted within asymmetric anticlines that have accumulated hydrocarbons, as in the Sacha, Shushufindi, and Auca fields. Production success and water injection-based secondary recovery programs, initiated in the 1980s, underscore its operational and economic relevance.

The Oriente Basin constitutes the most prolific segment of the Putumayo-Oriente-Marañón petroleum province. By 2010, it held an original oil-in-place volume of approximately 33 billion barrels across 125 fields. As of 2011, cumulative production totaled 4.6 billion barrels, corresponding to a recovery factor of 14% (Ribadeneira & Baby, 1999).

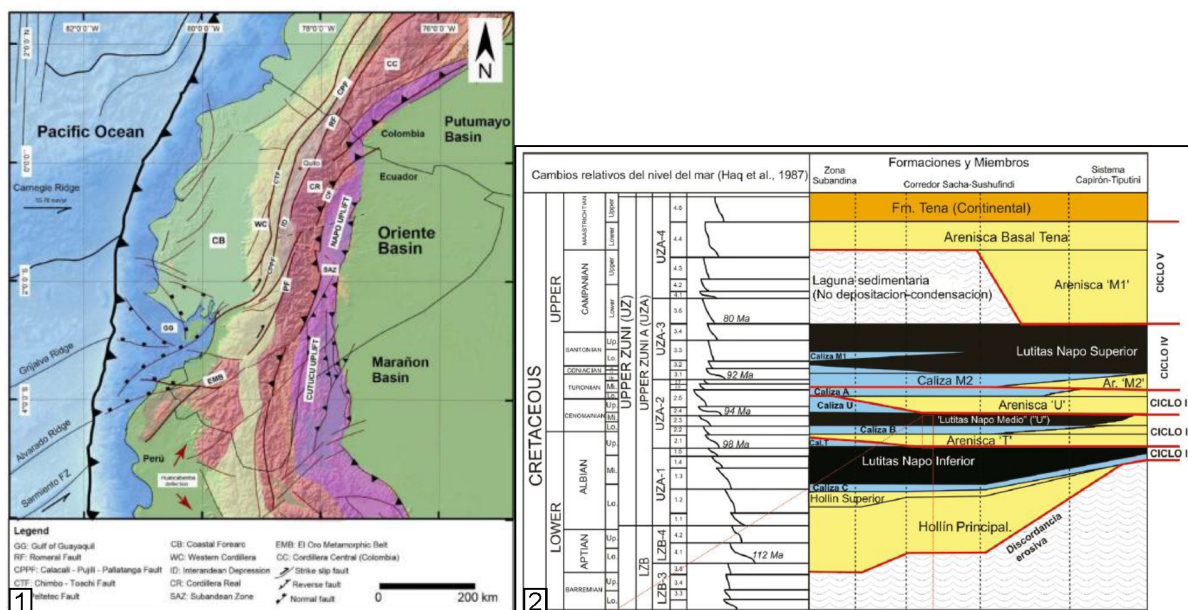


Figure 2—1) Schematic tectonic map of Ecuador showing the geological context of the Oriente Basin. Map adapted from Vallejo et al. (2021). 2) Formations, members, and sedimentary cycles of the Cretaceous of the Oriente Basin, and their relationship with the eustatic curve of Haq et al. (1987).

METHODOLOGY

To rigorously characterize the damage sensitivity of one of the main reservoirs of the Napo Formation, an experimental laboratory protocol was designed consisting of five interrelated stages. The workflow begins with a 60-foot continuous core extracted from one of the formation's principal sandy intervals. The core was subdivided into twenty cylindrical segments, each approximately 3 feet in length, stored in properly sealed aluminum tubes. Upon arrival at the laboratory—and following an initial verification of physical integrity and fluid preservation—each stage of the workflow applies progressively higher-resolution techniques, ranging from macroscopic analyses to microscopic characterizations. This multiscale approach aims to refine the diagnosis of damage mechanisms and to optimize mitigation strategies.

Stage 1

In this phase, the core is preserved intact within its aluminum tube to maintain its original saturation state. The process begins with a high-resolution computed tomography (CT) scan, which enables the identification of fracturing, density variations, lithological distribution, and diagenetic features within the core. Subsequently, a continuous gamma-ray log is acquired along each tube and correlated with the wireline log to calibrate depths and validate the stratigraphic framework.

The combination of these two techniques allows for a precise understanding and characterization of the extracted reservoir interval, even while the rock remains sealed inside the tubes—thereby preserving its native fluids.

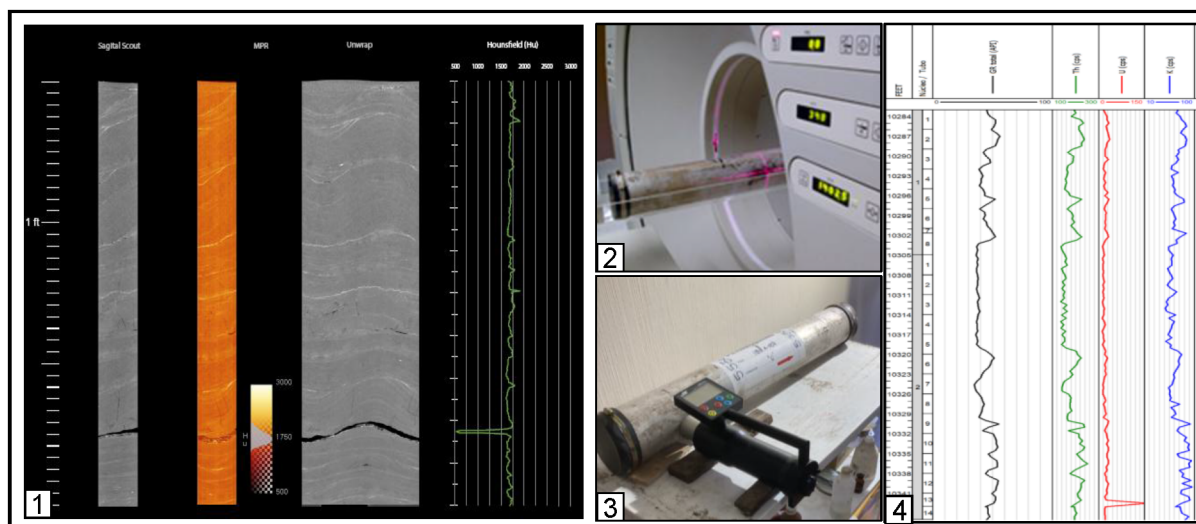


Figure 3—1) Example of processed CT scan from a 3-foot core section. 2) CT scanner. 3) Portable gamma-ray device. 4) Total and spectral gamma-ray curve over a 60-foot core interval.

Stage 2

The main focus of this stage is the analysis of basic petrophysical properties through the extraction of core plugs, selected based on the CT scan results and refined by a preliminary description of the interval. Sampling must be representative of the entire interval, aiming to characterize both the best and poorest reservoir-quality zones—thus avoiding overestimation of reservoir potential. Plugs are extracted with a length of 6 to 7 cm and a diameter of 38 mm. Both horizontally oriented plugs (parallel to bedding) and vertically oriented plugs (perpendicular to bedding) are obtained. The samples are preserved using film, aluminum foil, and paraffin. Several horizontal plugs are reserved for use in subsequent stages.

The results obtained during this stage include density, water saturation, oil saturation, porosity, and permeability (under standard and NOBP conditions) for each sample selected for basic petrophysical analysis.

Once the plugs have been extracted, the core is split longitudinally into two parts. One portion (referred to as $\frac{1}{3}$) is used to take high-resolution digital photographs under white light and ultraviolet light, providing a visual record of the study interval. This is followed by a sedimentological description of the interval to define sedimentary facies, facies associations, and interpret the depositional environment. Additionally, a probe-type minipermeameter is used to rapidly obtain preliminary permeability values directly on the remaining portion of the core (designated as $\frac{2}{3}$), prior to the full results of the basic petrophysical tests.

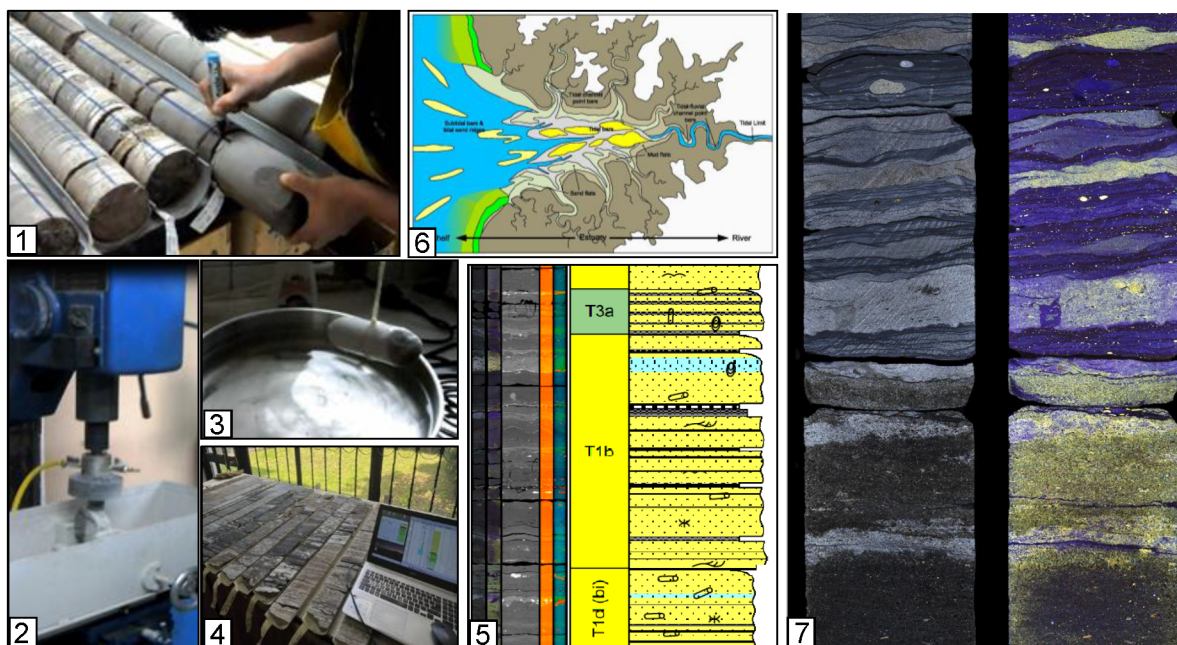


Figure 4—1) Core fitting and marking. 2) Plug extraction. 4-5) Example of sedimentological description. 6) Environmental interpretation. 7) High-resolution photograph under white light (left) and ultraviolet light (right).

Stage 3

This stage involves compositional, textural, and pore system characterization at multiple observational scales. Based on the basic petrophysical analysis, representative samples are selected for study through petrography, X-ray diffraction (XRD), and scanning electron microscopy (SEM). These analyses allow for a more detailed understanding of the reservoir.

Petrographic analysis focuses on grain size distributions, the composition and arrangement of rock components, and the identification of diagenetic processes. XRD is used to determine the mineralogical composition of the rock, with particular emphasis on identifying the types of clay minerals present. Finally, SEM provides insights into the pore system, enabling characterization of both pore sizes and pore throat dimensions.

Other analyses—such as geomechanics, geochemistry, and biostratigraphy—are not addressed in this paper but are nonetheless relevant to a comprehensive reservoir characterization.

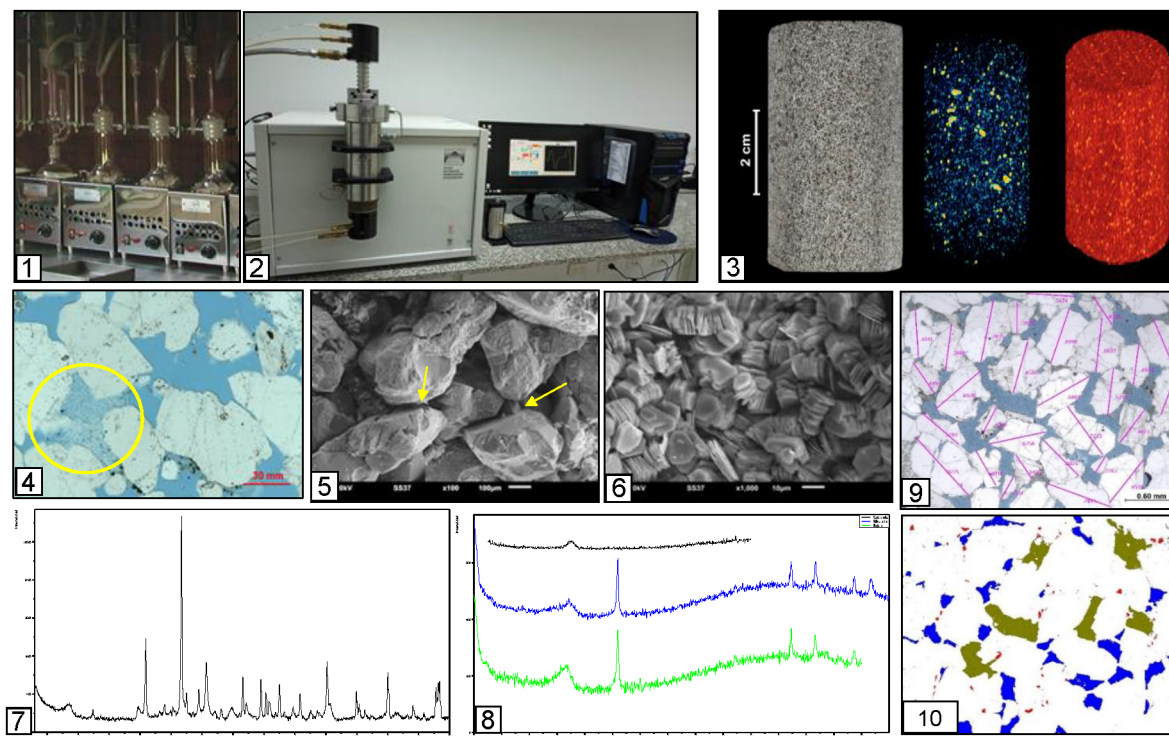


Figure 5—1)Hydrocarbon washing. 2) Basic petrophysical measurement equipment. 3) Washed plug photograph and CT scan. 4) Petrographic microphotograph. 5-6) High-resolution SEM images. 7) X-ray diffraction (XRD) pattern of whole rock. 8) XRD pattern of clay fraction. 9-10) Image analysis: grain size distribution and porosity.

Stage 4

Based on the integration of results obtained in Stages 2 and 3, a special core analysis (SCAL) program is proposed. This consists of a targeted set of advanced petrophysical studies conducted on a representative sample set selected according to basic petrophysical properties and the petrographic and mineralogical variability identified previously.

This stage includes wettability tests (Amott index, to determine the rock's affinity for water or oil), electrical properties measurements (formation factor and resistivity index, used to calibrate wireline logs: a, m, and n parameters), capillary pressure by mercury injection (to obtain the pore throat size distribution), and relative permeability curve measurements (to understand and predict multiphase fluid flow—such as water and oil—through the porous medium). These experiments reveal the rock's fluid-phase affinities and provide essential input curves for simulating reservoir dynamic behavior.

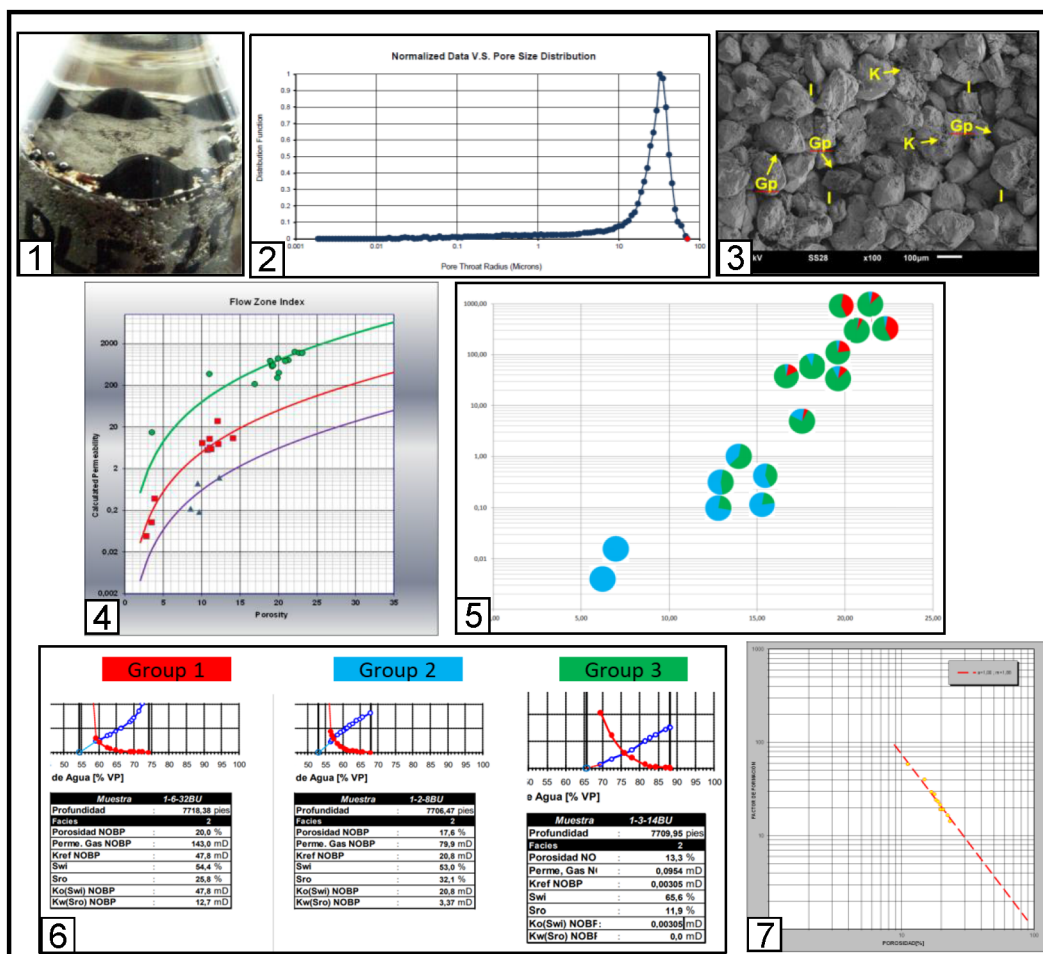


Figure 6—1) Wettability. 2) Pore throat size distribution. 3) Pore throats under SEM. 4) FZI curves. 5) Porosity-permeability cross-plots. 6) Relative permeability curves. 7) Formation factor.

Stage 5

This final stage represents the core objective of the present study. It aims to evaluate the consequences of introducing non-native fluids into the reservoir by replicating fluid invasion scenarios at laboratory scale—whether through water injection or the introduction of operational fluids—such as those used in drilling, completion, and hydraulic fracturing. The goal is to quantify and predict the degree of petrophysical alteration caused by these processes.

As in Stage 4, the selection of samples ensures representativity in terms of facies, petrophysical behavior, and the reservoir interval of interest. While Stages 4 and 5 are sometimes considered part of the same workflow, they are distinguished here to emphasize the specific formation damage testing component.

It is important to note that the reservoir under study is oil-bearing. Two workflows are presented: one focused on water analysis and the other on the evaluation of chemical product impact.

The first sequence involves the assessment of injection water to determine the effects of its interaction with the reservoir and to establish the minimum quality required to avoid adverse changes in porous media properties. This includes evaluating whether the water source will be purge water, another origin, or a blend. A compatibility test is conducted between formation water and aliquots of the intended injection water. Subsequently, critical salinity is assessed—particularly relevant when the injection water is fresher than the formation water. At the same time, critical velocity is determined. It is advisable to accompany this assessment with scanning electron microscopy (SEM) before and after testing (ensuring the sample has already been characterized in Stage 3) to identify potential fines migration or pore structure rearrangement.

Evaluating the filtration quality of the injection water is also critical to define the minimum acceptable filtration level that minimizes formation damage. Additionally, operational variants allow the assessment of suspended oil content — commonly present in purge water—when such water is used as injection fluid. These tests should also be supported by pre- and post-SEM analyses.

Another key element is assessing water sensitivity, which refers to the impact on reservoir permeability from the introduction of water with a composition different from that of the formation. This applies both to enhanced recovery operations and reservoir pressure maintenance programs.

Injectivity tests are conducted to evaluate the immediate effect of injection water on the reservoir's capacity to accept water without inducing damage in the injection zone. Damage is confirmed when increased pressure is required to maintain the same injection rate.

In addition, it is essential to evaluate the potential damage caused by the introduction of treatment products. Drilling, completion, fracturing, cementing, and general intervention fluids can lead to severe formation damage, potentially resulting in the abandonment of the well.

In this study, relative permeability curves (oil-water system) were used as the primary reference, both before and after product application. The analysis focused on curve shapes, crossover points, residual oil saturation, endpoint effective permeabilities, and production expressed as a percentage of OOIP, all serving as indicators of the impact of the tested control fluids.

A Relative Permeability Modifier (RPM) fluid was also evaluated, aimed at reducing effective water permeability to minimize the volume of produced water requiring disposal.

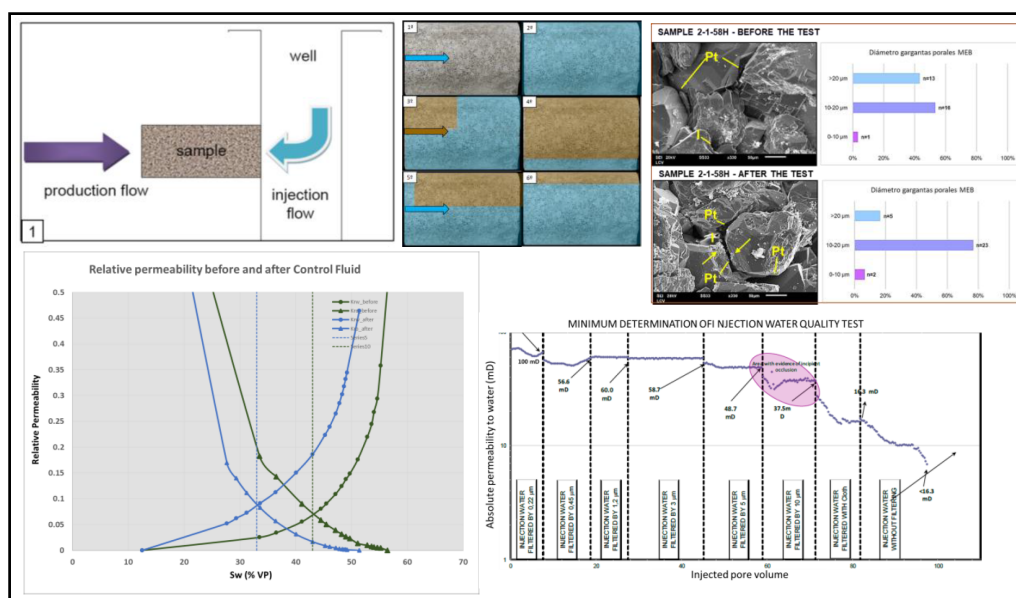


Figure 7—1) Plug test schematic. 2) SEM analysis, before and after testing. 3) Schematic of two-phase flow in plugs. 4) Relative permeability curves before and after control fluid application. 5) Water quality analysis.

RESULTS

Formation damage tests revealed up to a 40% loss in permeability in core samples after exposure to specific control fluids, mainly associated with changes in wettability and solid retention.

When broken down by fluid type, the variation in oil recovery (expressed as a percentage of OOIP) ranged from +3.6% to -39.5% for the control fluid, and from +0.22% to -14.7% for the fracturing fluid.

These results were made possible thanks to the availability of pre- and post-treatment data derived from relative permeability tests conducted before and after each injection scenario.

The RPM fluid partially fulfilled its objective by reducing effective water permeability by 16%, although this came at the expense of a 14% reduction in effective oil permeability.

In water injection zones, it was demonstrated that variations in the salinity of the injected water significantly impact reservoir productivity, leading to the definition of new quality thresholds for injection fluids.

CONCLUSIONS

The implementation of this experimental workflow provides a solid scientific basis for evaluating formation damage through laboratory testing. It enables operators to design flow protocols and targeted chemical treatments that minimize permeability loss and maximize hydrocarbon recovery. The integration of data from Stages 1 through 4 served as a foundation for comprehensive reservoir characterization, which, in Stage 5, was leveraged to reduce uncertainty in the selection of drilling, completion, and injection fluids. This framework offers a robust reference for the optimized management of mature conventional reservoirs.

Based on the laboratory findings, several key mitigation measures were proposed: optimized selection of bridging materials aligned with lithological characteristics, reformulation of fluids to preserve wettability, and the implementation of overbalanced drilling techniques using low-damage fluids.

This case study in the Napo Formation exemplifies the approach, demonstrating how dynamic behaviour observed in the laboratory can guide field operations. The structured workflow facilitates informed reservoir management decisions from the early stages of development and helps reduce uncertainty associated with productivity losses due to formation damage.

Reference

- Al-Yousef, H., Al-Qahtani, M., Al-Mutairi, S. et al 2015. Laboratory Investigation and Field Applications of Formation Damage Removal Techniques in Sandstone Reservoirs. SPE Middle East Oil & Gas Show and Conference. SPE-176398-MS. <https://doi.org/10.2118/176398-MS>
- Al-Mahrooqi, S., Grattoni, C. A., Muggeridge, A. H. et al 2006. Wettability Alteration During Low Salinity Waterflooding: Evaluation by Corefloods and NMR. *SCA International Symposium*. SCA2006-39. Available at: <https://www.scaweb.org>
- Baby, P., Rivadeneira, M., Barragan, R. et al 1999. Structural and stratigraphic evolution of the Oriente foreland basin of Ecuador. *J South Am Earth Sci* **12**(5): 371–391. [https://doi.org/10.1016/S0895-9811\(99\)00029-2](https://doi.org/10.1016/S0895-9811(99)00029-2)
- Bart, S., Mitchell, J., Smith, K. et al 1997. Let's get the most out of the existing wells. *Oilfield Review*, **9**(4), 2–21. Schlumberger.
- Bennion, D. B., & Thomas, F. B. 2005. Formation Damage: Fundamentals, Diagnosis and Mitigation. *Journal of Canadian Petroleum Technology*, **44**(11), 51–62. <https://doi.org/10.2118/02-11-04>
- Haq, B.U., Hardenbol, J., Vail, P.R., 1987. Chronology of sea levels since the triassic. *Science*, **235**, 1156–1167.
- Khilar, K. C., & Fogler, H. S. 1998. Migration of Fines in Porous Media. Springer. <https://doi.org/10.1007/978-1-4615-5523-3>
- McPhee, C., Reed, J., & Zubizarreta, I. 2015. Core analysis: A best practice guide (Vol. 64). Elsevier B.V. [10.1016/B978-0-444-63533-4.09989-3](https://doi.org/10.1016/B978-0-444-63533-4.09989-3)
- Morton-Thompson, D., & Woods, A. M. 1993. Development geology reference manual (p. 548). American Association of Petroleum Geologists.
- Mucci, M. Z. 2023. Let's talk about formation damage. First edition.
- Naguib, M. A., Alhanati, F. J., Schubarth, S. K. 2000. Guideline of artificial lift selection for mature fields. SPE Paper 64428. *Society of Petroleum Engineers*. [10.2118/64428-MS](https://doi.org/10.2118/64428-MS)
- Vallejo, C., Romero, C., Horton, B. et al 2021. Jurassic to Early Paleogene sedimentation in the Amazon region of Ecuador: Implications for the paleogeographic evolution of northwestern South America. *Global and Planetary Change*, **204**, 103555. <https://doi.org/10.1016/j.gloplacha.2021.103555>